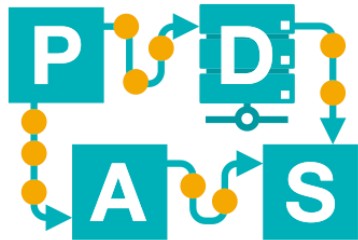


Association Rules and Sequence Mining

Instruction 8

IDS



Chair of Process
and Data Science

RWTHAACHEN
UNIVERSITY

Association Rules



Recap

Association rule mining can be seen as two steps:

1. Find all frequent item sets:

- Apriori algorithm
- FP-Growth algorithm

2. Generate strong association rules from the frequent item sets:

- By definition, these rules must satisfy minimum support and minimum confidence.

Why FP-Growth algorithm?

Disadvantages of Apriori algorithm:

- Find candidate sets in an expensive way (If frequent items are large in amount, so the combination would be huge and it would be an expensive operation.)
- Scan the database Repeatedly.

So Apriori algorithm is a slow algorithm.

FP-Growth algorithm

TID	Item sets
1	{1,2,3,4,5,6}
2	{7,2,3,4,5,6}
3	{1,8,4,5}
4	{1,9,10,4,6}
5	{10,2,2,4,11,5}

Item	Support count
1	3
2	3
3	2
4	5
5	4
6	3
7	1
8	1
9	1
10	2
11	1

FP-Growth algorithm

Min-support =3

Item	Support account
1	3
2	3
4	5
5	4
6	3

Write in
descending
order



L

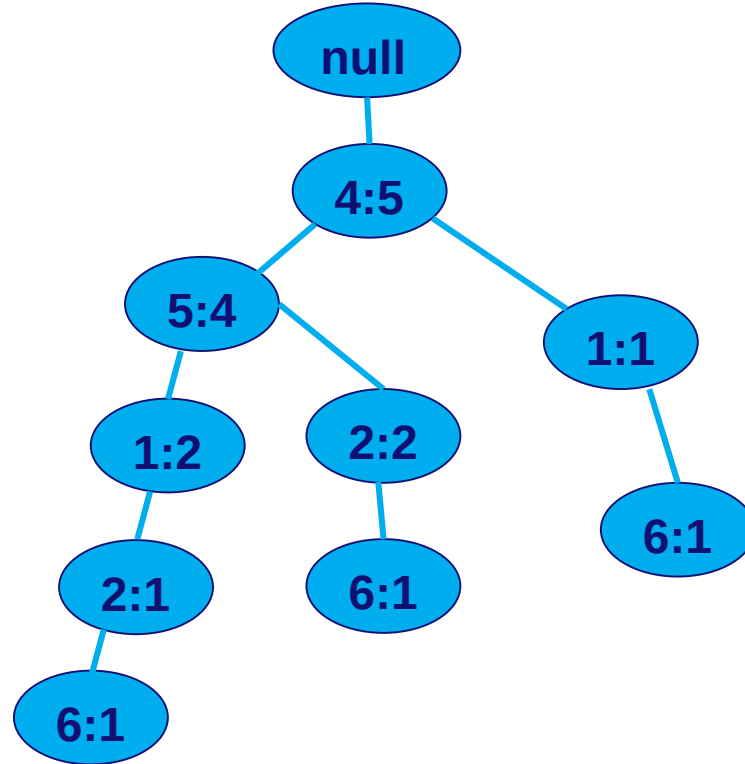
item	Support account
4	5
5	4
1	3
2	3
6	3

FP-Growth algorithm

TID	Item sets	Ordered item set
1	{1,2,3,4,5,6}	{4,5,1,2,6}
2	{7,2,3,4,5,6}	{4,5,2,6}
3	{1,8,4,5}	{4,5,1}
4	{1,9,10,4,6}	{4,1,6}
5	{10,2,2,4,11,5}	{4,5,2}

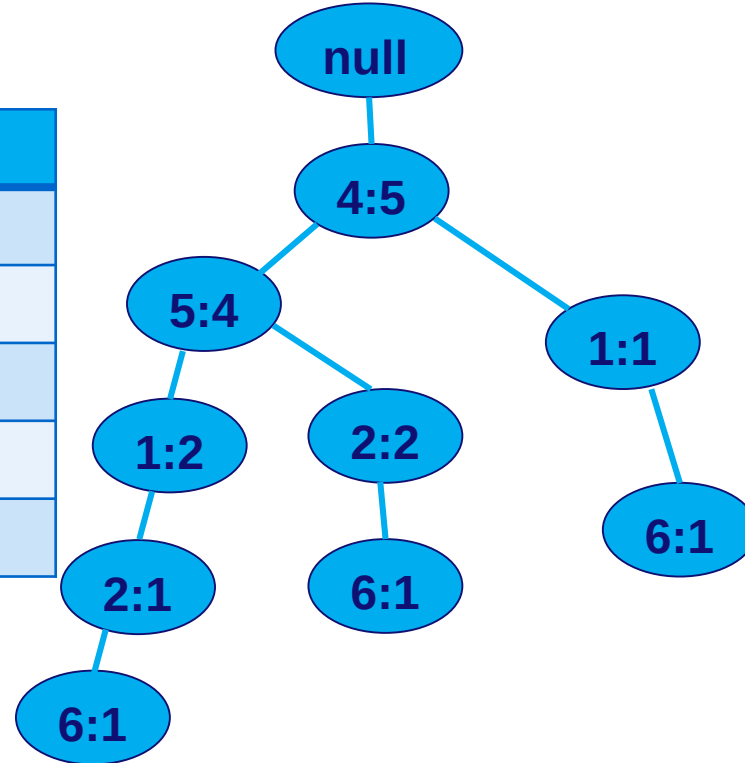
FP-Growth algorithm

TID	Ordered item set
1	{4,5,1,2,6}
2	{4,5,2,6}
3	{4,5,1}
4	{4,1,6}
5	{4,5,2}



FP-Growth algorithm

Item	Conditional pattern base
6	({4,5,1,2}:1, {4,5,2}:1,{4,1}:1)
2	({4,5,1}:1, {4,5}:2)
1	({4,5}:2, {4}:1)
5	({4}: 4)
4	-



FP-Growth algorithm

Item	Conditional pattern base	Conditional FP tree	Frequent pattern generated
6	(({4,5,1,2}:1, {4,5,2}:1,{4,1}:1)	[4:3]	<4,6: 3>
2	(({4,5,1}:1, {4,5}:2)	[4,5:3]	<4,2:3> <5,2:3><2,5,4:3>
1	(({4,5}:2, {4}:1)	[4:3]	<4,1:3>
5	(({4}: 4)	[4:4]	<4,5:4>
4	-	-	-

Next step: association rules... what is confidence and support for $4 \Rightarrow 6$ and $6 \Rightarrow 4$?

Association Rules

Set **min_sup** to 0.5, **min_conf** to 0.7. Is **{bread} => {meat}** from **D** a desired association rule?

TID	Set of items
0	bread, meat, wine
1	bread, meat
2	pizza, wine
3	bread, meat, pizza, wine

Set of transactions D

Association Rules

Set **min_sup** to 0.5, **min_conf** to 0.7. Is **{bread} => {meat}** from **D** a desired association rule?

TID	Set of items
0	bread, meat, wine
1	bread, meat
2	pizza, wine
3	bread, meat, pizza, wine

Set of transactions **D**

$$\text{support}(\{\text{bread}\} \Rightarrow \{\text{meat}\}) = \frac{\text{support}_{\text{count}}(\{\text{bread, meat}\})}{|D|} = \frac{3}{4} = 0.75 > \text{min_sup}$$

$$\text{confidence}(\{\text{bread}\} \Rightarrow \{\text{meat}\}) = \frac{\text{support}_{\text{count}}(\{\text{bread, meat}\})}{\text{support}_{\text{count}}(\{\text{bread}\})} = \frac{3}{3} = 1 > \text{min_conf}$$



{bread} => {meat} is a desired association rule

Association Rules

We use **lift** to evaluate the quality of the discovered association rule **A => B**

$$\text{lift}(A \Rightarrow B) = \frac{\text{support}(A \cup B)}{\text{support}(A) \cdot \text{support}(B)} = \frac{P(A \cup B)}{P(A) \cdot P(B)}$$

If $\text{lift}(A \Rightarrow B) \approx 1$ then A and B are independent

If $\text{lift}(A \Rightarrow B) \ll 1$ then A and B are negatively correlated

If $\text{lift}(A \Rightarrow B) \gg 1$ then A and B are positively correlated

Association Rules

Evaluate the quality of the association rule **{bread} => {meat}** by using **lift**:

TID	Set of items
0	bread, meat, wine
1	bread, meat
2	pizza, wine
3	bread, meat, pizza, wine

Set of transactions D

$$\text{lift}(\{bread\} \Rightarrow \{meat\}) = \frac{\text{support}(\{bread, meat\})}{\text{support}(\{bread\}) \cdot \text{support}(\{meat\})} = \frac{(3/4)}{(3/4) \cdot (3/4)} = 1.33$$

Association Rules

Exercise 3: Judge if $\{A, B\} \Rightarrow \{E\}$, $\{A\} \Rightarrow \{B\}$ and $\{A\} \Rightarrow \{C\}$ are the desired association rules under minimum support 0.5 and minimum confidence 0.75? Also evaluate the quality of the desired rules.

TID	Data items
1	A, B, E
2	C, A, D
3	C, B, D
4	C, A, B, E

Example data set S

Association Rules

- **Solution 3: Judge if $\{A, B\} \Rightarrow \{E\}$, $\{A\} \Rightarrow \{B\}$ and $\{A\} \Rightarrow \{C\}$**
- **$\text{support}(\{A, B\} \Rightarrow \{E\}) = 0.5$, $\text{confidence}(\{A, B\} \Rightarrow \{E\}) = 1$, $\text{lift}(\{A, B\} \Rightarrow \{E\}) = 2$, it is a desired association rule, and lift is larger than 1**
- **$\text{support}(\{A\} \Rightarrow \{B\}) = 0.5$, $\text{confidence}(\{A\} \Rightarrow \{B\}) = 0.67$, $\text{lift}(\{A\} \Rightarrow \{B\}) = 0.89$, it is not a desired association rule**
- **$\text{support}(\{A\} \Rightarrow \{C\}) = 0.5$, $\text{confidence}(\{A\} \Rightarrow \{C\}) = 0.67$, $\text{lift}(\{A\} \Rightarrow \{C\}) = 0.89$, it is not a desired association rule**

Sequence Mining



Support

In this database, find the support count of:

(bc)(de)

b(de)

(ac)(bc)

(bc)(ac)

D = [
<a(bc)d(eb)>,
<(ac)(bc)de>,
<(ac)b(cd)>,
<(bc)(bd)(bde)>,
<(abc)(ac)(bc)de>,
<a(bd)c(de)>
]

Support: solutions

In this database, find the support of:

(bc)(de): **1**

b(de)

(ac)(bc)

(bc)(ac)

D = [
<a(bc)d(eb)>,
<(ac)(bc)de>,
<(ac)b(cd)>,
<(bc)(bd)(bde)>,
<(abc)(ac)(bc)de>,
<a(bd)c(de)>
]

Support: solutions

In this database, find the support of:

(bc)(de): **1**

b(de): **2**

(ac)(bc)

(bc)(ac)

D = [
<a(bc)d(eb)>,
<(ac)(bc)de>,
<(ac)b(cd)>,
<(bc)(bd)(bde)>,
<(abc)(ac)(bc)de>,
<a(bd)c(de)>
]

Support: solutions

In this database, find the support of:

(bc)(de): **1**

b(de): **2**

(ac)(bc): **2**

(bc)(ac)

D = [
<a(bc)d(eb)>,
<(ac)(bc)de>,
<(ac)b(cd)>,
<(bc)(bd)(bde)>,
<(abc)(ac)(bc)de>,
<a(bd)c(de)>
]

Support: solutions

In this database, find the support of:

(bc)(de): **1**

b(de): **2**

(ac)(bc): **2**

(bc)(ac): **1**

D = [
<a(bc)d(eb)>,
<(ac)(bc)de>,
<(ac)b(cd)>,
<(bc)(bd)(bde)>,
<(abc)(ac)(bc)de>,
<a(bd)c(de)>
]